

Subject Description Form

Subject Code	ITC331FC
Subject Title	3D CAD Pattern on Ladies Wear
Credit Value	3
Level	3
Pre-requisite/ <Co-requisite>/ (Exclusion)	ITC211FC Digital Product Creation or ITC313XC Fashion Product Ideation and Development (ITC3208T 3D CAD Pattern)
Objectives	The subject provides the knowledge of application of 3D CAD on apparel pattern design. It trains the use of 3D CAD techniques in virtual garment assembly. It also explores 3D CAD technique and support in all stages of the apparel product development process, including pattern design and garment styling.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: a. understand 3D CAD system in pattern design; b. demonstrate the ability of using 3D CAD technologies in apparel product development, including 3D avatar customization and body fitting; c. develop 3D patterns by virtual garment assembly; d. make use of fabric images and properties in 3D simulation; e. demonstrate the outfit of garment by virtual cat walk simulation; and f. examine the limitations of existing 3D CAD technologies and systems
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	(I) Overview the virtual garment assembly General sequence of garment assembly Method of garment assembly (II) 3D material Calibration and identification of fabric properties (III) 3D CAD pattern design 3D pattern tracing Application of 3D CAD on style alternation/ modifications. (IV) 3D styling Simulation of dress up sequences and poses (V) 3D simulation and cat walk presentation 3D rendering Cat walk runway Stage setting and lighting control

Teaching/Learning Methodology <i>(Note 3)</i>	Studio sessions will be used to deliver teaching in this subject with more interactive teaching and learning, and problem solving. Hands-on experience of CAD and demonstrations of pattern design techniques will also be provided. Online learning materials and self-learning exercises with step-by-step instructions will also be incorporated via LEARN@PolyU to enhance students’ understanding of apparel construction.									
Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	Specific assessment methods/tasks		% weighting		Intended subject learning outcomes to be assessed					
					a	b	c	d	e	f
	1. In-class exercises		40%		✓	✓	✓	✓	✓	✓
	2. Projects		60%		✓	✓	✓	✓	✓	✓
	Total		100 %							
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Students will be expected to understand basic techniques of clothing CAD pattern design in this subject. Such techniques will be acquired through extensive hands-on practice and exercises. Therefore, the course will be assessed by in-class assignments. Students will also be expected to communicate effectively and think critically by applying clothing CAD to support their future career, and such knowledge integration will be achieved through project work. In-class assignments and projects will be appropriate methods to assess students’ proficiency in achieving the intended learning outcomes.										
Student Study Effort Expected	Class contact:									
	▪ Studio					38 Hrs.				
	Other student study effort:									
	▪ Self-study/Preparation					20 Hrs.				
	▪ Project/Assignments					50 Hrs.				
	Total student study effort					108 Hrs.				
Reading List and References	<u>Books</u>									
	Faerm, S. (2010). Fashion Design Course: Principles, Practice and Techniques: The Ultimate Guide For Aspiring Fashion Designers. Lontoo: Thames & Hudson.									
	Fashionary International Ltd. (2017), Fashionpedia: The visual dictionary of fashion design.									
	Nugent, L. (2009), Computerized Patternmaking for Apparel Production. Fairchild Books, US.									
	Santamaria, G. (2014), Men in this town: London, Tokyo, Sydney, Milan, New York. Hardie Grant Books, Richmond, Victoria.									
Werle, S. (2009). Fashionista: a century of style icons. Prestel Pub.										
<u>Websites</u>										
CLO 3D, https://www.clo3d.com/										